Lesson 7: Dendroclimatology - handout

Reading Environment and History through Tree Rings / 90 minutes / Draft / 30 October 2026

Define the reconstruction target

Dendroclimatology relates tree-ring measurements to instrumental climate and estimates earlier conditions. Secure calendar dating does not automatically establish accurate climate estimates. Define region, season, variable and units before interpreting a curve.

Biological conditions

Responses depend on limiting temperature or moisture, species, soils and crown position. Stored resources and buds can carry previous-year effects into later growth. Consider seasons and mechanisms rather than only a correlation with annual averages. Competition, insects, pollution and management remain relevant alternatives from Lesson 6.

Chronology and instrumental observations

Check boundaries, measurements and crossdating. Distinguish cores from independent trees and report annual sample depth, stability of the common signal and weaker early replication. For climate observations, examine station location, elevation, terrain, missing data, observational changes and adjustments.

Detrending and timescale

A flexible reference curve can remove slow climatic variation as well as age trends. A long chronology assembled from short individual series can face limits in recovering slow variation, discussed as the segment length curse. Test whether processing retains the variation required by the question.

Exercise A: calibration and verification

8 minutes: calculate 4, compare 3, record 1. All data are synthetic and do not demonstrate the accuracy of a real reconstruction.

Calibration uses 2001–2004. Ring indices I are 0.7, 0.9, 1.1 and 1.3; temperatures are 8.9, 9.5, 10.3 and 11.3° C. These teaching values give the equation T_est = 6 + 4I. I is dimensionless and T is a seasonal mean temperature in ° C. Do not transfer the coefficients to another forest.

Verification uses 2005–2007. Hold the equation and selection procedure fixed when evaluating unused observations. Four and three years are chosen for arithmetic, not recommended research lengths.

Year	Index I	Observed T (° C)	Estimate / error
2005	0.8	9.0	Complete
2006	1.0	10.1	Complete
2007	1.2	10.9	Complete

Calculations and questions

1. Calculate each prediction and error T_est − T.

2. Does zero mean error imply zero error in every year?

3. Explain which observations were used to estimate coefficients.

4. What happens if the test period is used to change the chosen season or processing?

Season, variable and processing searches are part of model building. Once test results guide those choices, that period has contributed to selection. Use new independent evidence or a design evaluating the complete selection procedure.

Exercise B: compare correlation and error

6 minutes. Model A is T_est = 6 + 4I. Model B adds 1° C to every A estimate. Compare both with a constant prediction of 10° C.

Year	Observed (° C)	A (° C)	B (° C)	Baseline (° C)
2005	9.0	9.2	10.2	10.0
2006	10.1	10.0	11.0	10.0
2007	10.9	10.8	11.8	10.0

Calculation

For root mean squared error, square the errors, average them and take the square root. The unit here is ° C. Work through those three steps for each prediction.

Interpretation

1. Correlation measures shared variation.

2. Bias is systematic over- or underestimation.

3. Adding a constant preserves correlation but changes prediction error.

4. Check improvement over a simple constant prediction too.

Do not generalize real-model performance from these three years. Research also examines autocorrelation, searching across candidates and changing seasonal or temporal responses. Mathematical supplement BG03 develops the equations further.

A Japanese case and response stability

Hinoki and early spring temperature

Yonenobu and Eckstein (2006) examined February–April temperature in central Japan using Hinoki ring widths. Values below are reported in Table 1, not recalculated for this course.

Stage	Period	Correlation r
Calibration	1900–1961	0.514
Verification	1962–2001	0.078

The later loss of predictive skill illustrates why a successful calibration need not remain valid indefinitely. The paper discusses possible explanations including pollution, without establishing one cause for every mismatch. Compare observations, forest history and other proxies.

Beyond width

Nakatsuka and colleagues (2020) integrate oxygen and hydrogen isotope information to examine summer climate in central Japan. Physiology and tree age still affect isotopes; they are not direct rainfall measurements.

Reading questions

Which region and season are represented? Which periods were used for calibration and independent verification? Are unsuccessful periods reported? Is a proposed cause a hypothesis or a conclusion supported by independent evidence?

- R52 Early spring temperatures from Hinoki cypress - Yonenobu & Eckstein (2006)
- R53 Integrating oxygen and hydrogen isotopes - Nakatsuka et al. (2020)

Uncertainty and the scope of a conclusion

Read the figure

The 1800–1810 slide curve is synthetic. Its bounds add and subtract a chosen 0.8° C and are not statistical 95% intervals. In real figures, check the interval's meaning, calculation, assumptions and omissions.

Separate uncertainty sources

Residual variation, coefficient estimation, observational error, chronology quality, declining replication, response change and processing choices are distinct issues. Secure calendar dating does not eliminate uncertainty in the climate variable.

Region, season and extremes

One site's spring temperature differs from global annual temperature. Test spatial correspondence with observations. Outside the calibration range, consider growth limits and absent rings. Identifying an extreme year differs from estimating its magnitude.

Retain records

Keep data, identifiers, units, missing values, settings, equations, seasonal choices, time splits and reasons for exclusions.

Exit questions

1. Explain how high correlation can coexist with bias.
2. Why independently evaluate season selection as well as coefficients?
3. What information must accompany a reconstruction curve?

Terms and further reading

Term	Brief explanation
dendroclimatology	Estimating past climate using tested tree-ring relationships.
calibration	Estimating a model from paired observations.
verification	Evaluating against information unused in building the model.
bias	Systematic over- or underestimation.
autocorrelation	Association between time-separated values in one series.
extrapolation	Estimation beyond the range over which a model was tested.

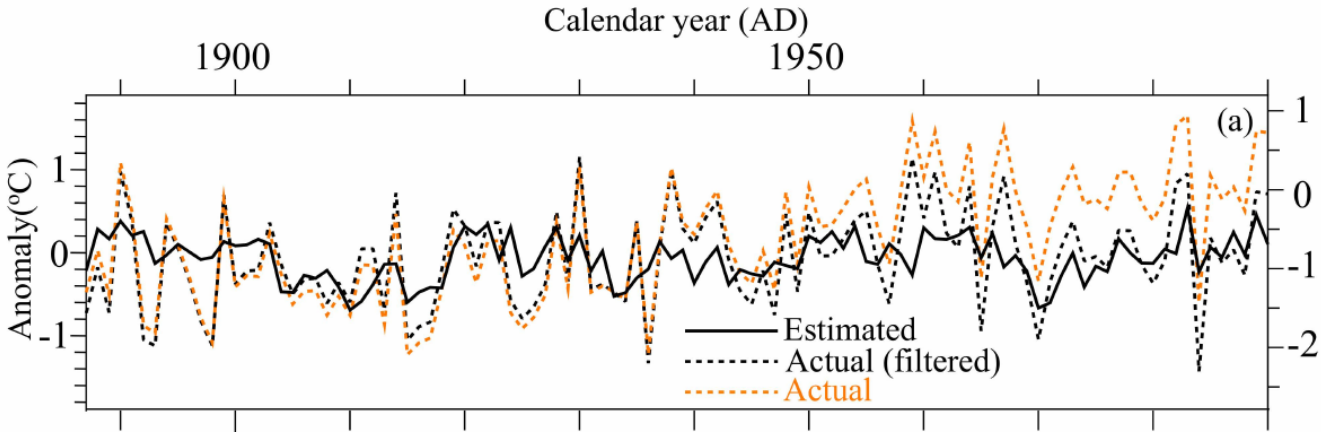
- R08 International Tree-Ring Data Bank - NOAA NCEI
- R52 Early spring temperatures from Hinoki cypress - Yonenobu & Eckstein (2006)
- R53 Integrating oxygen and hydrogen isotopes - Nakatsuka et al. (2020)
- R54 The segment length curse - Cook et al. (1995)

Foundational reading: Speer (2010), Chapter 8; Fritts (1976), Chapters 5–8. Supplement BG03 develops mathematical definitions and evaluation measures.

Review a conclusion and its limits, checking variable, season and region. Lesson 8 combines archaeology, ecology and climate in a research plan. Consult the syllabus guidance of 2–4 A4 pages and at least three references for the final report.

Lesson 7 Read the axes and legend in an original figure

For slide 27. Use this enlargement in print. Ohyama et al. (2013), Figure 3(a), concerns March–May temperature in northeastern Asia.



Panel (a), cropped from the original. Horizontal axis: calendar year. Both vertical axes use °C, but represent differently processed series and different scales.

Ohyama et al., 2013, Fig. 3(a). CC BY 3.0.

Line	Axis and meaning
Solid black	Left: estimated temperature
Dotted black	Left: filtered observations
Dotted orange	Right: unfiltered observations

Identify the appropriate axis before comparing values. Does agreement in this panel establish predictive performance for unseen years?

This is the 1887–1990 calibration fit, not an independent validation plot. Vertical separation of lines on different axes is not directly a temperature difference.

Ohyama et al. (2013), Fig. 3(a), CC BY 3.0

Lesson 7 What does CE compare?

In one sentence

The coefficient of efficiency, CE, measures the reduction in squared prediction error relative to predicting the verification-period observed mean in every year.

$CE = 1 - \Sigma(\text{observed} - \text{predicted})^2 / \Sigma(\text{observed} - \text{verification mean})^2$

CE	Interpretation
1	Perfect predictions for these verification data
0	Same squared error as the verification-mean benchmark
Negative	Larger squared error than that benchmark

The mean is a retrospective benchmark used to assess predictions; it is not assumed to be known when forecasting unseen years. The expression is undefined if the observed verification series is constant.

Table 2 reports split-period CE values of 0.019 and −0.040. Do both support a stable reconstruction? Distinguish the roles of Figure 3(a) and Table 2.

Check

The first represents only a small improvement over the benchmark; the second performs worse. A convincing calibration plot cannot substitute for independent testing. Consider periods, processing and validation design. The processed series in this study also cannot directly recover the long-term warming trend.

Ohyama et al. (2013), Table 2